Description of ISU-ILCC Battery Aging Dataset

1. General Information

1.1. Data title: ISU-ILCC Battery Aging Dataset

1.2. Authors

- Tingkai Li
  Iowa State University (Present Affiliation: University of Connecticut)
  ORCID: 0000-0003-1934-1000
  Emails: tkl@iastate.edu; tingkai.li@uconn.edu
- Jinqiang Liu
  Iowa State University
  ORCID: 0000-0002-1106-6473
  Email: jinqiang@iastate.edu
- Adam Thelen
  Iowa State University
  ORCID: 0000-0001-5443-3678
  Email: acthelen@iastate.edu
- Chad Tischer
  Iowa Lakes Community College
  Email: ctischer@iowalakes.edu
- Chao Hu (Corresponding author)
  Iowa State University (Present Affiliation: University of Connecticut)
  ORCID: 0000-0001-9228-7675
  Emails: chaohu@iastate.edu; chao.hu@uconn.edu

1.3. Associated publication:

Predicting Battery Lifetime Under Varying Usage Conditions from Early Aging Data

(The preprint is available on arXiv at: https://doi.org/10.48550/arXiv.2307.08382)

2. Dataset Overview

The ISU-ILCC battery aging dataset was collected jointly by the System Reliability and Safety Laboratory at Iowa State University (ISU), now the Reliability Engineering and Informatics Laboratory (REIL) at the University of Connecticut, and Iowa Lakes Community College (ILCC). The dataset is designed to study the dependency of battery capacity fade from three stress factors: charge rate, discharge rate, and depth of discharge. The dataset contains cycle aging data from 251 lithium-ion (Li-ion) polymer cells (also called lithium polymer cells) cycled under 63 unique conditions.

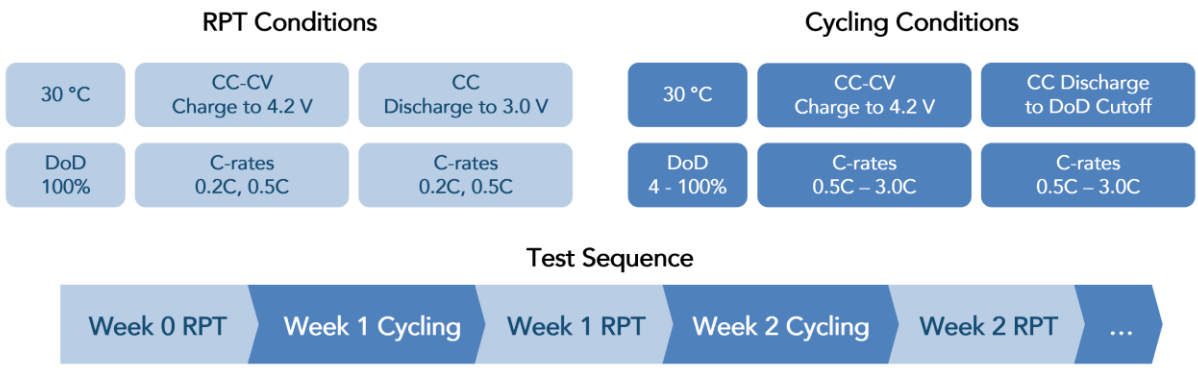


Figure 1: Summary of test conditions and sequence.

3. Experiment Design

3.1. Cell specification

The battery cells used in this aging test campaign are 502030 Li-ion polymer cells manufactured by Shenzhen Honghaosheng Electronics Co., Ltd. in Shenzhen, China. These cells have lithium nickel manganese cobalt oxide (NMC) as the cathode active material and graphite as the anode active material. The rated capacity is 250 mAh, and the operating voltage window is between 3.0 V and 4.2 V.

3.2. Test equipment

The cells were cycle-aged using two 64-channel Neware BTS-4000 battery cyclers, one located at ISU and the other one at ILCC. Cells' terminals were clipped to individual cell holders and placed inside two Thermo Scientific™ incubators set at a constant temperature of 30 °C. Figure 2 shows the equipment setup at ILCC, which is also similar to the one at ISU.

Figure 2: Equipment setup at ILCC.

3.3. Regular cycling test

The regular cycling test was designed with three varying stress factors: charging C-rate, discharging C-rate, and depths of discharge (DoD). One charge/discharge cycle consisted of a constant-current/constant-voltage (CC/CV) charging step and a constant-voltage discharging step. The depths of discharge for different groups were determined by estimating the voltage cutoffs during the CC discharging step. The cycling profiles are shown in Figure 3.

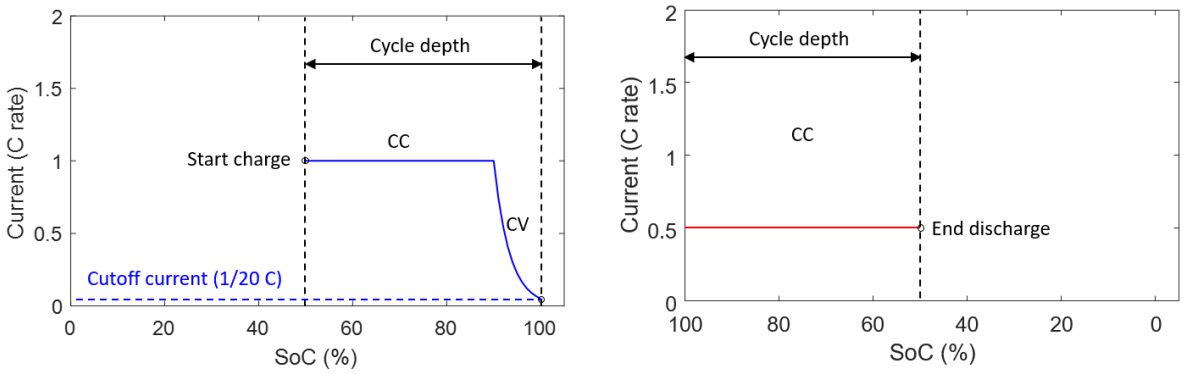


Figure 3: Charge and discharge profiles for the cycling test.

3.4.Reference performance test

A reference performance test (RPT) was periodically run on every cell to measure the full discharge capacity and collect data for lifetime prediction. Using an RPT to check cell capacity is important for measuring the full capacity of cells cycled with partial DoDs. During the RPT, two full cycles of CC/CV charging and CC discharging at different cycling rates were performed, one at C/5 and the other at C/2. The RPT discharge capacity at C/5 is used in generating capacity fade plots. The voltage-capacity data from a complete C/5 discharge cycle are used for feature extraction lifetime prediction. Additionally, for groups 30–64, we added a cycle with pulse tests at different SoC levels to obtain the DC internal resistances. The RPT profiles are shown in Figure 4.

The RPT is performed after 7 days of cycling. Groups 20 through 64 have an additional RPT after the first 3.5 days of cycling (i.e., after the first half week of cycling) to better capture the initial capacity fade.

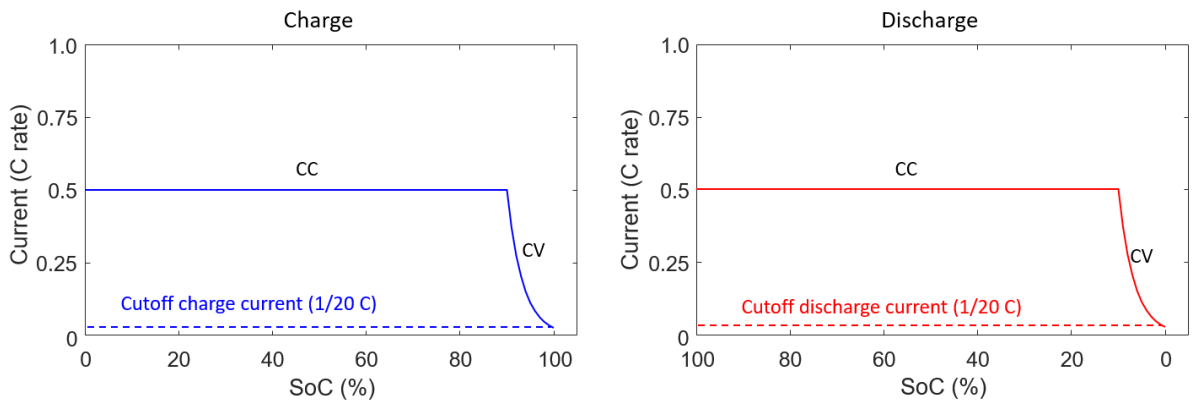


Figure 4: Charge and discharge profiles for the RPT.

3.5. Cell ID

Each cell was assigned a unique identifier (i.e., cell ID) that consisted of the cell's group number and the cell's index within the group. For example. G3C1 is the first cell in group three.

4. Data Structure

In the published dataset, each cell has two separate data files, one for the RPT data and the other for the cycling data. All data files are processed using Python and saved in JSON format. Files are named by the cell ID. To convert the data to Python dictionaries, a sample code has been provided with this dataset ("load_data.py").

* The 0.5-week RPT is only for groups 20–64.

Notation	**Description**	**Unit**
Capacity_discharge Capacity_charge	Total Capacity discharged or charged of a specific cycle.	Ah
start_stop_time (RPT)	Start and stop time of a specific RPT. (For calculating time on test)	YYYY-MM-DD hh:mm:ss
start_stop_time (Cycling)	Start and stop time of a specific week's cycling. (For calculating time on test)	YYYY-MM-DD hh:mm:ss
time_series_charge time_series_discharge	The start and stop times of each cycle's charging and discharging steps.	YYYY-MM-DD hh:mm:ss
Q	Charge recordings during a specific cycle.	Ah
V	Voltage recordings during a specific cycle.	V
I	Current recordings during a specific cycle.	A
E	Energy recordings during a specific cycle.	Wh
t	Time stamps of recordings during a specific cycle	YYYY-MM-DD hh:mm:ss

4.1. File directory

1. RPT_json [folder] – *Raw data*
   1.1. G1C1.json … G64C4.json – RPT data for each cell in JSON format.
2. Cycling_json [folder] – *Raw data*
   2.1. G1C1.json … G64C4.json – Cycling data for each cell in JSON format.
3. Q_interpolated [folder] – *Processed data*
   3.1. G1C1.csv … G64C4.csv – Interpolated Q data with respect to a voltage vector containing 1000 elements equally spaced between 3V and 4.2V. Each column contains the Q reading from one RPT.
4. capacity_fade [folder] – *Processed data*
   4.1. G1C1.csv … G64C4.csv – Two-column tabular data for capacity fade curves of each cell. The first column is the time on test, and the second column is the remaining capacity.

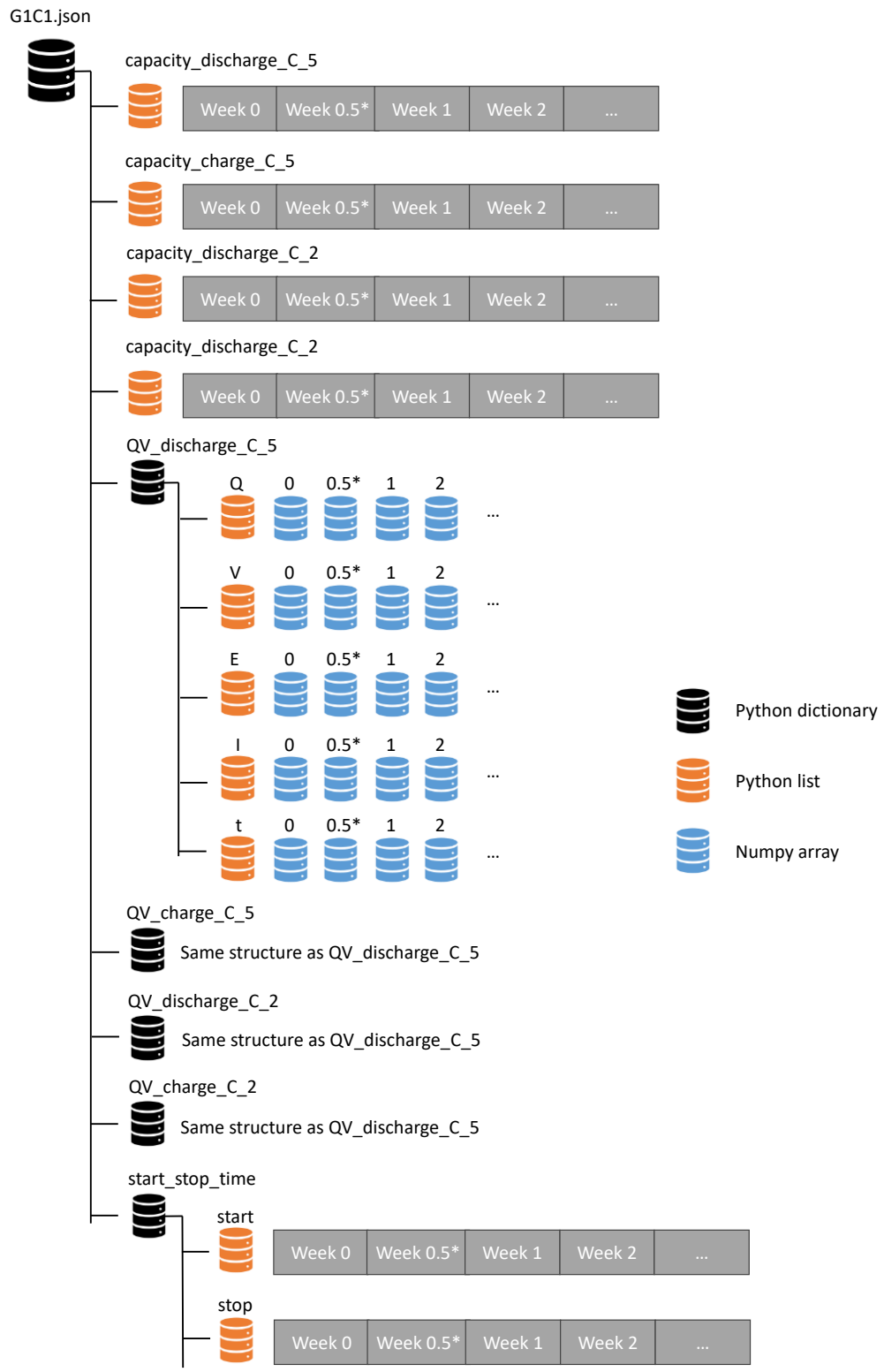
5. Valid_cells.csv – CSV file contains the Cell ID for all the cells published in the current release.
6. process_data.py – A sample code containing functions converting the RPT and the cycling JSON files to Python dictionaries. Also, this sample code provided processing functions to obtain the data for interpolated Q-V curves and capacity fade curves.

   This code is developed in the following environment:

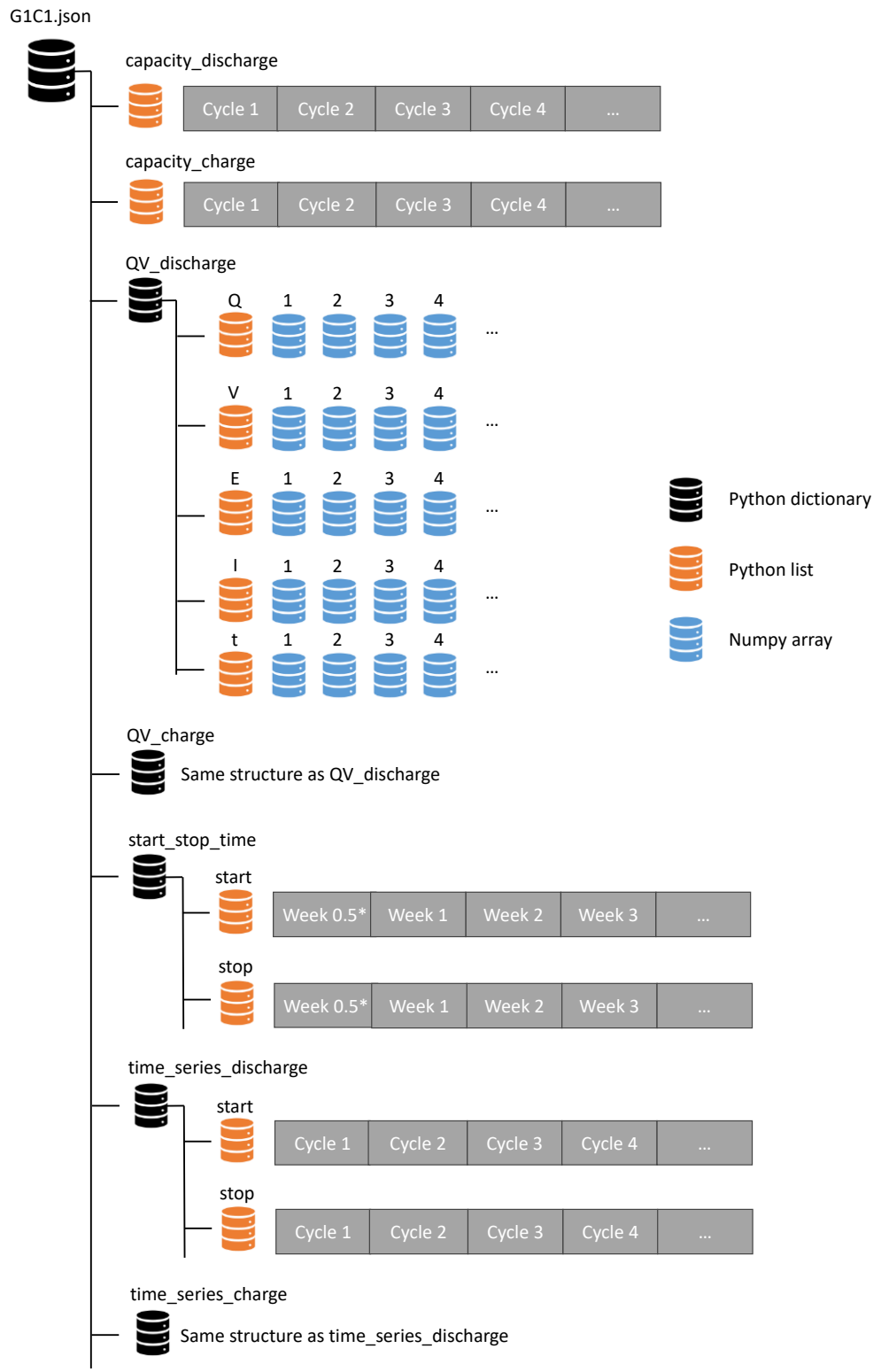
   - python: 3.7.13 (Version 3.9.9 is compatible with packages below)
   - pandas: 1.3.5
   - scipy: 1.7.3
   - numpy: 1.21.5

   Note: An interpolation warning is triggered on line 164 for several cells when the assigned smoothing factor ("precision_C_5") is too low. Without intervention, this would result in an array of NaN for the variable "tck". To resolve this problem, lines 165-167 loops to correct the smoothing factor until the variable "tck" is no longer an array of NaN to ensure correct output. Therefore, you may ignore the warning.

4.2. Structure of RPT data



4.3. Structure of cycling data



5. Acknowledgment

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7. Update for This Release

- The dataset of 251 cells is fully released.
- Table A1 is revised to include cycling conditions for G11.

Appendix

A1. Cycling conditions

Group #	Charging C-rate	Discharging C-rate	Mean DoD
1	0.500	0.500	25.90%
2	0.500	0.500	46.11%
3	0.500	0.500	67.77%
4	1.000	0.500	47.27%
5	1.000	0.500	97.54%
6	2.000	0.500	27.23%
7	2.000	0.500	46.86%
8	2.000	0.500	67.24%
9	2.000	0.500	96.93%
10	2.500	0.500	47.21%
11	2.500	0.500	95.98%
12	3.000	0.500	28.60%
13	3.000	0.500	47.26%
14	3.000	0.500	66.32%
16	0.500	0.500	97.62%
17	1.000	0.500	67.53%
18	2.500	0.500	27.53%
19	2.500	0.500	65.70%
20	0.800	0.500	98.16%
21	1.200	0.500	97.98%
22	1.400	0.500	97.88%
23	1.600	0.500	97.61%
24	1.800	0.500	97.51%
25	1.800	0.600	51.43%
26	1.400	2.200	4.16%
27	0.600	2.400	57.85%
28	2.400	1.600	80.99%
29	1.600	1.800	52.48%
30	0.800	0.800	77.59%
31	1.200	1.000	61.43%
32	1.000	1.400	38.86%
33	2.000	1.200	40.65%
34	2.200	2.000	27.60%
35	1.825	0.500	97.02%
36	2.075	0.500	97.06%
37	0.725	0.500	98.08%
38	1.875	0.500	95.83%
39	1.475	0.500	97.14%
40	1.825	1.025	31.94%
41	2.075	1.775	67.02%

Group #	Charging C-rate	Discharging C-rate	Mean DoD
42	0.725	2.375	36.32%
43	1.875	2.325	38.00%
44	0.775	1.275	37.12%
45	1.125	1.725	80.26%
46	1.225	2.025	75.36%
47	2.325	1.925	48.27%
48	2.375	2.225	58.57%
49	0.975	0.675	18.82%
50	2.425	1.625	20.10%
51	2.275	1.875	53.64%
52	1.425	0.875	83.55%
53	2.025	0.825	91.54%
54	0.925	1.125	27.05%
55	1.025	2.475	61.44%
56	2.175	0.975	53.81%
57	1.775	1.175	17.42%
58	2.475	0.575	42.03%
59	1.325	1.825	31.64%
60	0.675	1.325	83.63%
61	2.125	1.975	54.07%
62	1.575	2.425	64.19%
63	1.975	1.675	80.57%
64	1.175	1.425	91.44%

A2. Notes for known abnormalities

- For groups 20–29, we have unrecoverable data loss for the 4th week's RPTs. During the data processing, all of the data was replaced by empty Python lists.
- For cells G25C1, G26C1, G26C2, G26C3, and G26C4, we have unrecoverable data loss for the 23rd and 24th weeks' RPTs, and the 24th week's cycling tests.